

Inferring the 3D shapes of galaxies inside halos with UNIONS: preparation for intrinsic alignment studies

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Fabian Hervas Peters

Laboratory: CEA, Paris-Saclay, CosmoStat



UNIONS meeting, Paris

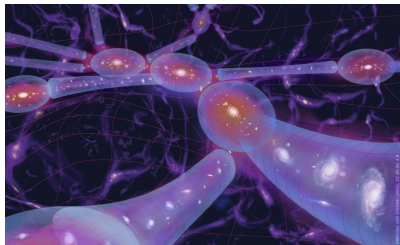
July 9th, 2026

<https://github.com/antonincorinaldi/antonincorinaldi.github.io/tree/main/Talks>

Context and objectives

Galaxy-halo connection for intrinsic alignments

- **Galaxy-halo connection:** the link between galaxies and halos is not clear

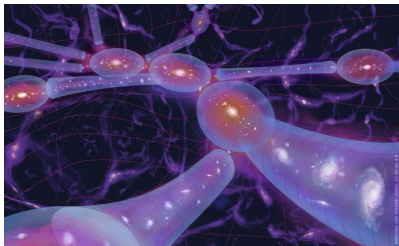


Credit: Fortuna & Chisari (2025)

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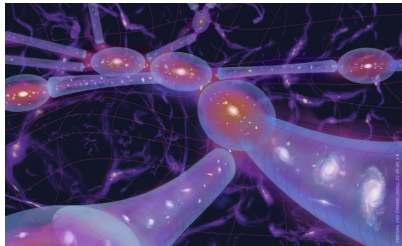


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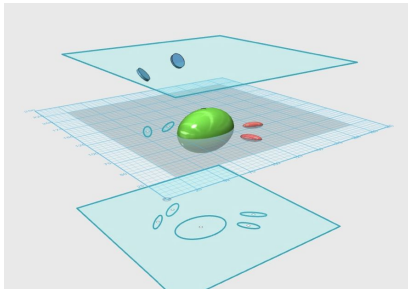
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- This is interesting for **intrinsic alignments** (IA): the IA signal depends on the way the **shapes** of galaxies evolved in 3D among the **large-scale structure** of the Universe



Joachimi et al. 2016

Context and objectives

- **Aims:**

- 1) Developing a model for the 3D morphology of galaxies inside dark matter halos using **N-body simulations**
- 2) Inferring the galaxy-halo connection and the distribution of the shapes of galaxies **in 3D** from their **observed projected images**

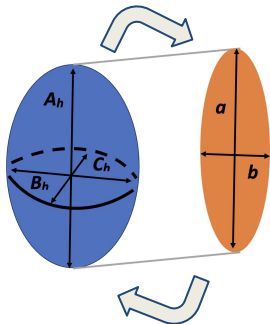
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- **Problem:**

=> How to go from projected images to 3D shapes ?



UNIONS (Ultraviolet Near Infrared Optical Northern Survey)

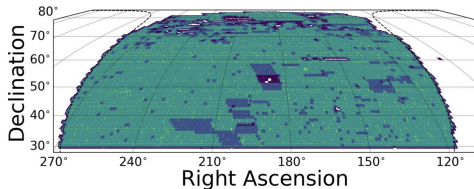


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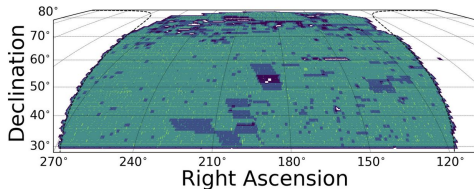


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~340 millions of galaxy shapes
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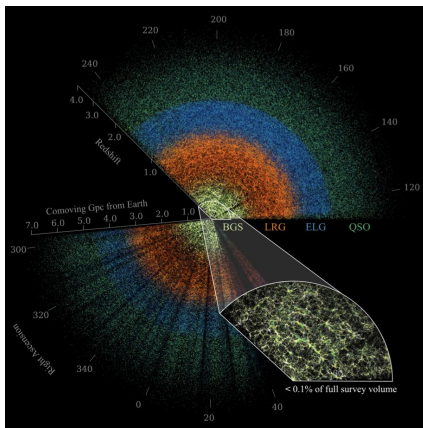
- **Measure of intrinsic alignments** by cross-matching UNIONS 2D shapes of galaxies with **spectroscopic** surveys ([Hervas Peters et al. 2025](#))

UNIONS x DESI

- Spectroscopic samples of galaxies:
 - **ELG**, **LRG** and **BGS**, Dark Energy Spectroscopic Instrument (DESI) DR1



Credit: DESI Legacy Imaging Survey



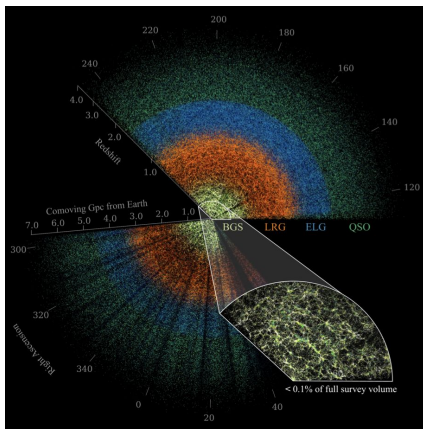
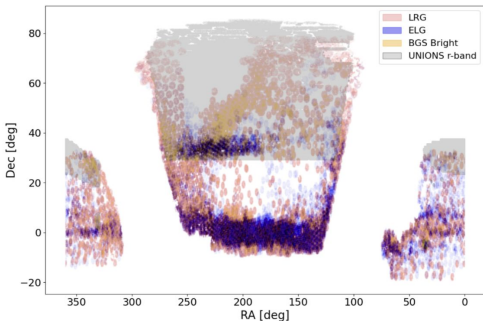
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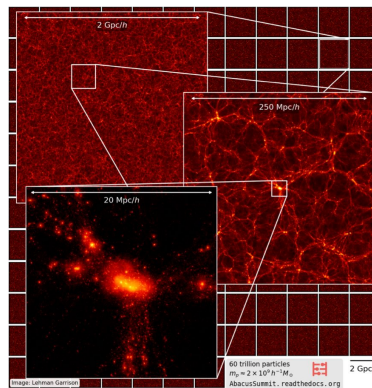


DESI Collaboration 2025

- Cross-match with **UNIONS** forms
 - . **ELG-UNIONS**: ~360 000 galaxies
 - . **LRG-UNIONS**: ~360 000 galaxies
 - . **BGS-UNIONS**: ~1 300 000 galaxies

N-body simulations AbacusSummit

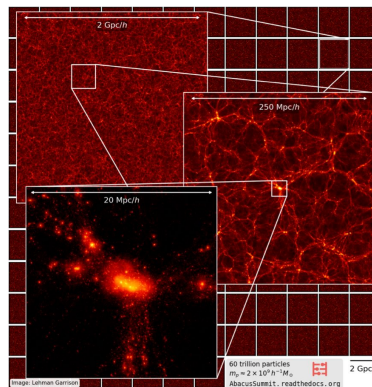
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Maksimova et al. 2021

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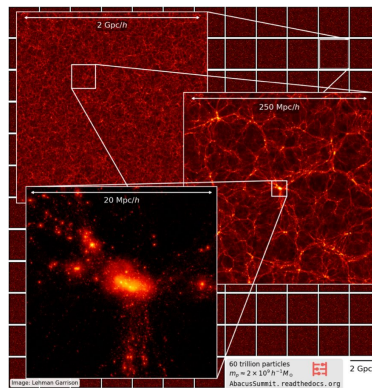
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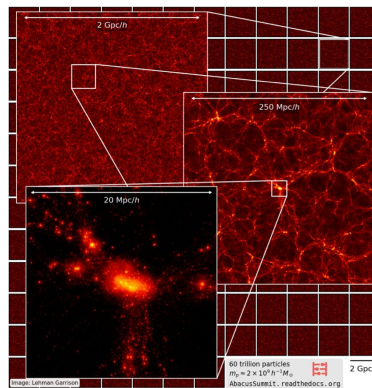
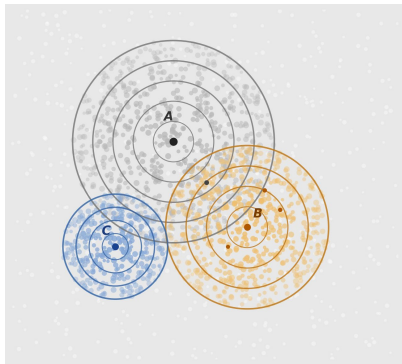
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- **Catalogues of halos** identified with the CompaSO (Competitive Assignment to Spherical Overdensities) algorithm (*Hadzhiyska et al. 2021*)

Methodology

I) Getting a halo catalogue per galaxy sample

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III) Modelling the 3D shapes of galaxies inside halos

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IV) Inferring galaxy-halo connection from observations
using simulation-based inference (SBI)

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- *Step 1:* generate a **mock galaxy catalogue** in a simulation box using an **HOD (Halo Occupation Distribution) model** for each galaxy sample

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Example: HOD model for the LRG ([Yuan et al. 2023](#))

Mean occupation number of **central** and **satellite** galaxies:

$$\bar{n}_{\text{cent}}^{\text{LRG}}(M) = \frac{f_{\text{ic}}}{2} \text{erfc} \left[\frac{\log_{10}(M_{\text{cut}}/M)}{\sqrt{2}\sigma} \right] \quad \bar{n}_{\text{sat}}^{\text{LRG}}(M) = \left[\frac{M - \kappa M_{\text{cut}}}{M_1} \right]^{\alpha} \bar{n}_{\text{cent}}^{\text{LRG}}(M)$$

- M mass of the host halo
 M_{cut} minimum halo mass to set a central galaxy
 f_{ic} incompleteness parameter (by definition: $0 < f_{\text{ic}} \leq 1$)
 σ steepness of the transition from 0 to 1 in the number of central
 κM_{cut} minimum halo mass to host a satellite galaxy
 α power law index on the number of satellite galaxies
 M_1 typical halo mass that hosts one satellite galaxy

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I) Getting a halo catalogue per galaxy sample

- Step 2: do **clustering measurements** with the data and the mock and check that both match well (*Landy & Szalay 1993*)

$$\xi_{gg}(r_p, \Pi) = \frac{DD(r_p, \Pi) - 2 DR(r_p, \Pi) + RR(r_p, \Pi)}{RR(r_p, \Pi)}$$

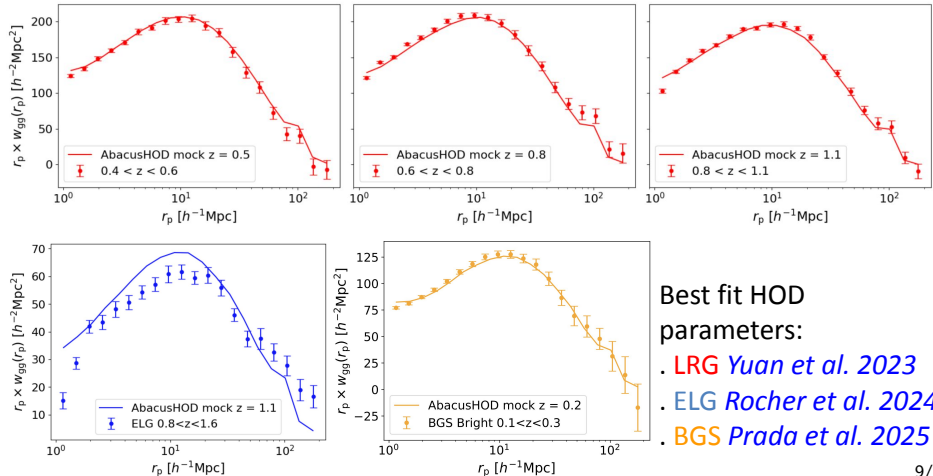
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Best fit HOD
parameters:

- **LRG** *Yuan et al. 2023*
- **ELG** *Rocher et al. 2024*
- **BGS** *Prada et al. 2025*

II) Modelling the 3D shapes of halos

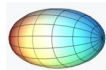
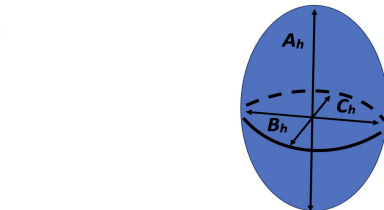
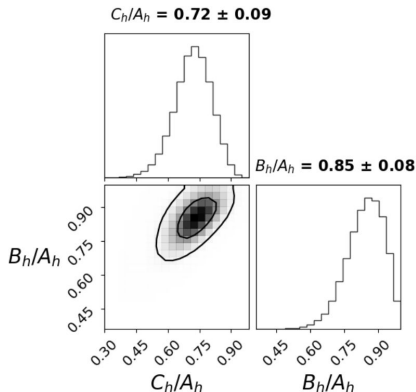
- Simple inertia tensor:
$$I_{ij} = \frac{\sum_n m_n x_{ni} x_{nj}}{\sum_n m_n}$$

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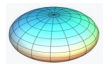
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prolate $A_h > B_h \approx C_h$



oblate $A_h \approx B_h > C_h$



spheroidal $A_h \approx B_h \approx C_h$

Example: Abacus halos at $z = 0.5$

Pandya et al. 2024

III) Modelling the 3D shapes of galaxies inside halos

- **Populating** each halo with a 3D **central galaxy** of axis lengths A_g, B_g, C_g such that:

$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h} \quad \text{with } \tau_B, \tau_C \in [0; 1]$$

Host halo
Galaxy

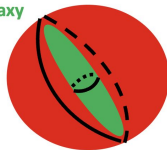


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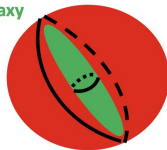
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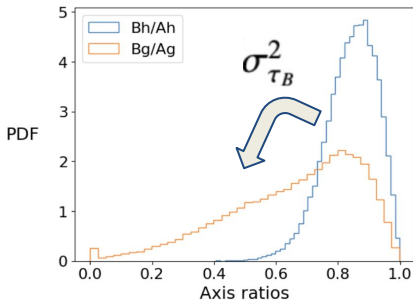
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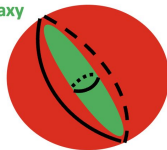
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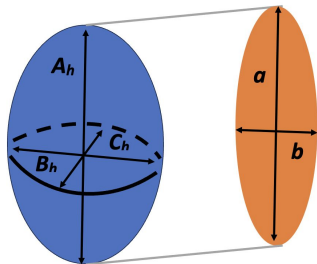
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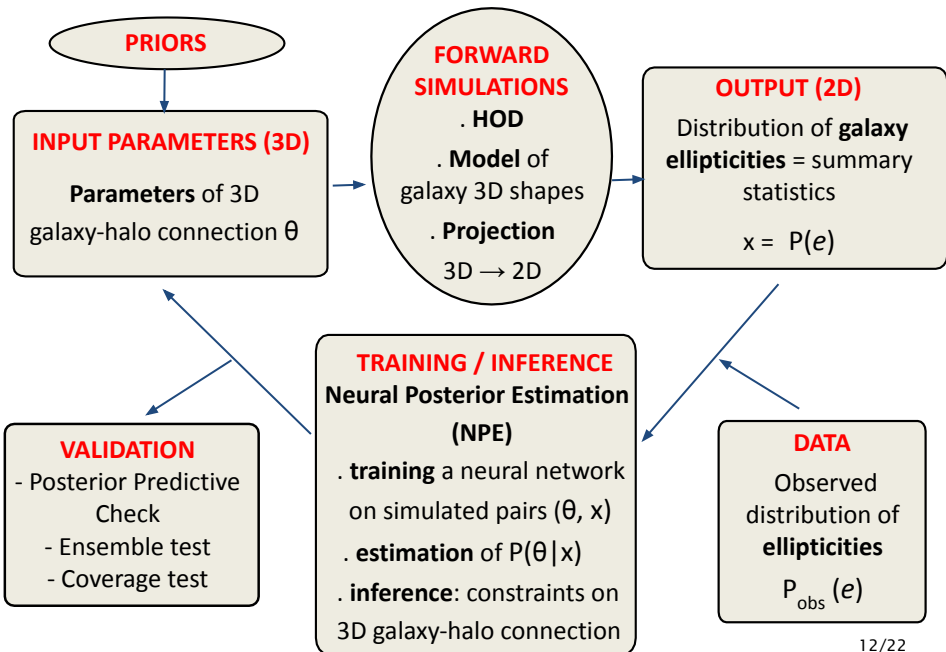
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- 4 free parameters: $\theta = \{\mu_{\tau_B}, \mu_{\tau_C}, \sigma_{\tau_B}, \sigma_{\tau_C}\}$
- **Projection** in 2D ([Lamman et al. 2023](#)) for different $\mu_{\tau_B}, \mu_{\tau_C}$ drawn uniformly in $[0; 1]$.



IV) Inferring 3D galaxy-halo connection with SBI



Analyses

- **(1)** LRG in 3 redshift bins

$$0.4 < z < 0.6$$

$$0.6 < z < 0.8$$

$$0.8 < z < 1.1$$

=> testing for redshift evolution

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- (2)** ELG $0.8 < z < 1.6$ and LRG $0.4 < z < 0.6$

=> testing for different types of galaxies

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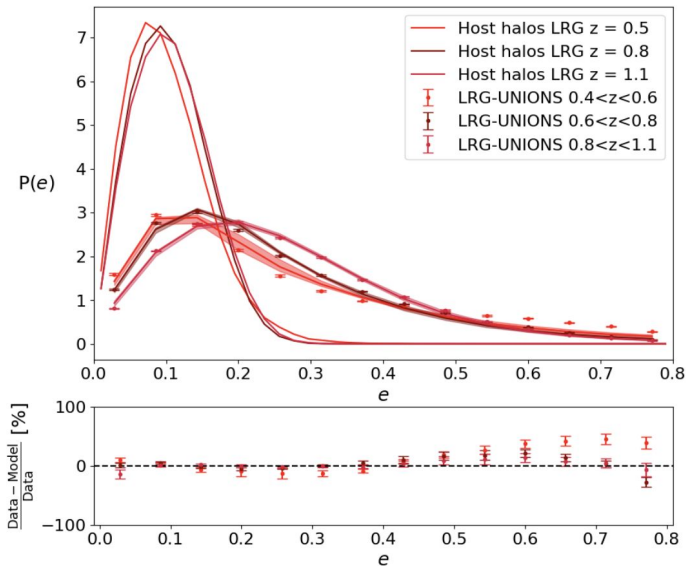
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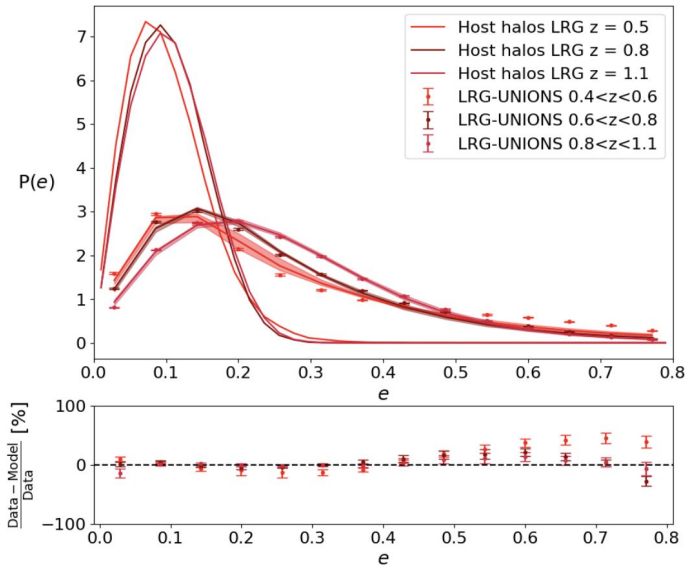
- For each case:
 - distribution of ellipticities
 - galaxy-halo connection
 - constraint on 3D shapes

(1) Redshift evolution: distribution of ellipticities



Corinaldi et al. 2026, in prep

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Corinaldi et al. 2026, in prep

LRG 2D shapes more elongated at higher redshift

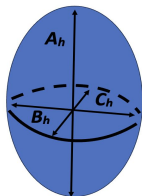
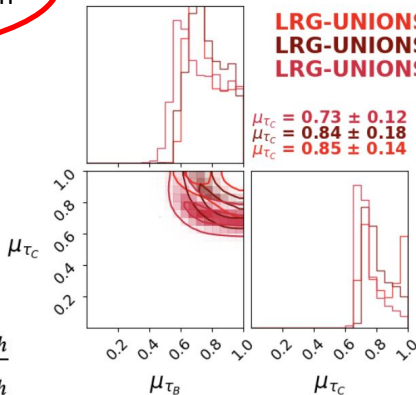
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$\mu_{\tau_B}, \mu_{\tau_C}$: means of τ_B, τ_C
for the whole halo population

$$\begin{aligned}\mu_{\tau_B} &= 0.72 \pm 0.18 \\ \mu_{\tau_B} &= 0.75 \pm 0.13 \\ \mu_{\tau_B} &= 0.85 \pm 0.22\end{aligned}$$

LRG-UNIONS $0.4 < z < 0.6$
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LRG-UNIONS $0.8 < z < 1.1$

$$\begin{aligned}\mu_{\tau_C} &= 0.73 \pm 0.12 \\ \mu_{\tau_C} &= 0.84 \pm 0.18 \\ \mu_{\tau_C} &= 0.85 \pm 0.14\end{aligned}$$



$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h}$$

with $\tau_B, \tau_C \in [0; 1]$

Corinaldi et al. 2026, in prep

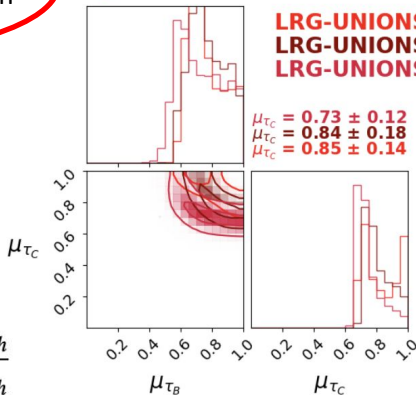
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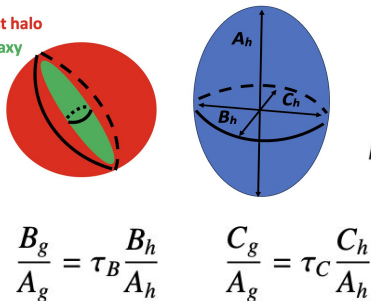
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LRG 3D shapes more stretched inside their halos at higher redshift

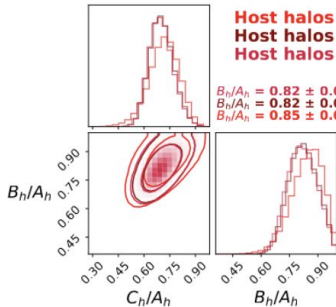
(1) Redshift evolution: constraints on 3D shapes

Halos

$$\begin{aligned}C_h/A_h &= 0.69 \pm 0.07 \\C_h/A_h &= 0.70 \pm 0.07 \\C_h/A_h &= 0.72 \pm 0.09\end{aligned}$$

Host halos LRG $z=0.5$
Host halos LRG $z=0.8$
Host halos LRG $z=1.1$

$$\begin{aligned}B_h/A_h &= 0.82 \pm 0.08 \\B_h/A_h &= 0.82 \pm 0.08 \\B_h/A_h &= 0.85 \pm 0.08\end{aligned}$$

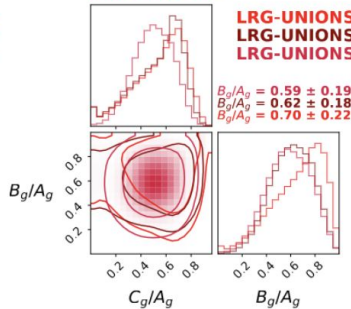


Galaxies

$$\begin{aligned}C_g/A_g &= 0.50 \pm 0.16 \\C_g/A_g &= 0.58 \pm 0.20 \\C_g/A_g &= 0.59 \pm 0.21\end{aligned}$$

LRG-UNIONS $0.4 < z < 0.6$
LRG-UNIONS $0.6 < z < 0.8$
LRG-UNIONS $0.8 < z < 1.1$

$$\begin{aligned}B_g/A_g &= 0.59 \pm 0.19 \\B_g/A_g &= 0.62 \pm 0.18 \\B_g/A_g &= 0.70 \pm 0.22\end{aligned}$$



Corinaldi et al. 2026, in prep

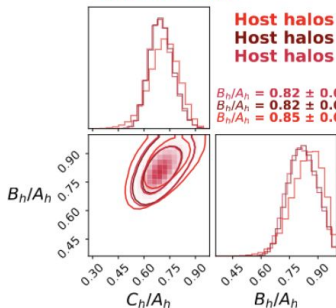
(1) Redshift evolution: constraints on 3D shapes

Halos

$$\begin{aligned}C_h/A_h &= 0.69 \pm 0.07 \\C_h/A_h &= 0.70 \pm 0.07 \\C_h/A_h &= 0.72 \pm 0.09\end{aligned}$$

Host halos LRG $z=0.5$
Host halos LRG $z=0.8$
Host halos LRG $z=1.1$

$$\begin{aligned}B_h/A_h &= 0.82 \pm 0.08 \\B_h/A_h &= 0.82 \pm 0.08 \\B_h/A_h &= 0.85 \pm 0.08\end{aligned}$$

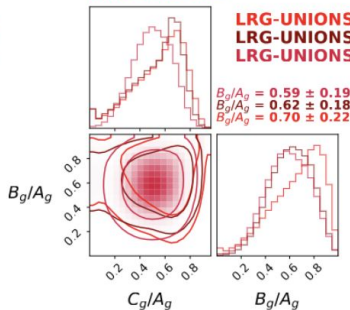


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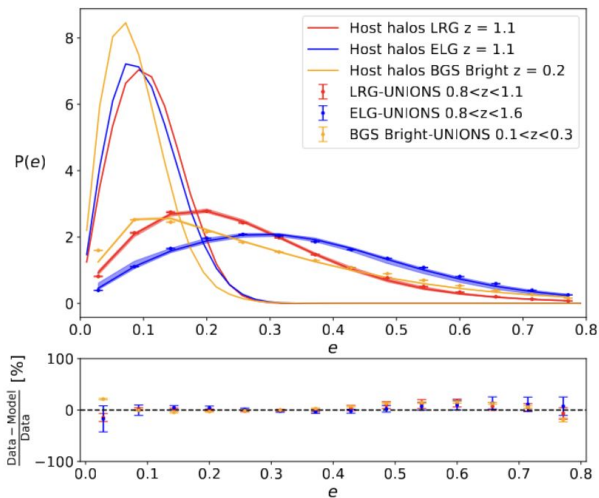
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Corinaldi et al. 2026, in prep

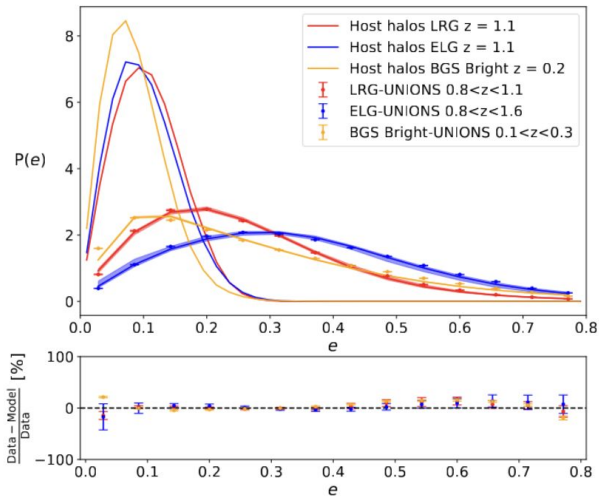
LRG 3D shapes more elongated at higher redshift
Host halos rounder than the galaxies

(2) ELG-LRG-BGS: distributions of ellipticities



Corinaldi et al. 2026, in prep

(2) ELG-LRG-BGS: distributions of ellipticities



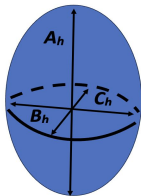
Corinaldi et al. 2026, in prep

ELG 2D shapes more scattered than LRG
BGS 2D shapes rounder than ELG and LRG

(2) ELG-LRG-BGS: galaxy-halo connection

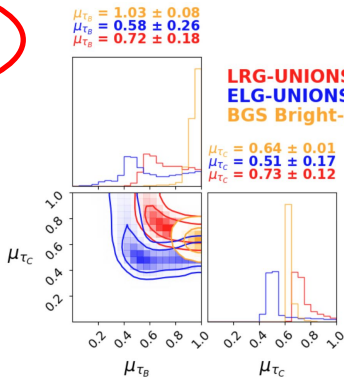
$\mu_{\tau_B}, \mu_{\tau_C}$: means of τ_B, τ_C
for the whole population

Host halo
Galaxy



$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h}$$

with $\tau_B, \tau_C \in [0; 1]$

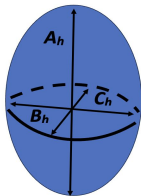


Corinaldi et al. 2026, in prep

(2) ELG-LRG-BGS: galaxy-halo connection

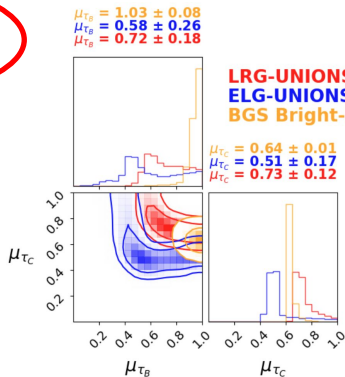
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with $\tau_B, \tau_C \in [0; 1]$



Corinaldi et al. 2026, in prep

We probe different 3D galaxy-halo connection between LRG, ELG and BGS

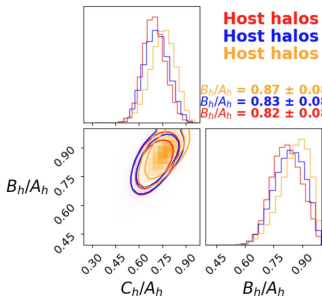
(2) ELG-LRG-BGS: constraints on 3D shapes

Halos

$$\begin{aligned} C_h/A_h &= 0.75 \pm 0.08 \\ C_h/A_h &= 0.71 \pm 0.08 \\ C_h/A_h &= 0.69 \pm 0.07 \end{aligned}$$

Host halos LRG $z = 1.1$
Host halos ELG $z = 1.1$
Host halos BGS BRIGHT $z = 0.2$

$$\begin{aligned} B_h/A_h &= 0.87 \pm 0.08 \\ B_h/A_h &= 0.83 \pm 0.08 \\ B_h/A_h &= 0.82 \pm 0.08 \end{aligned}$$

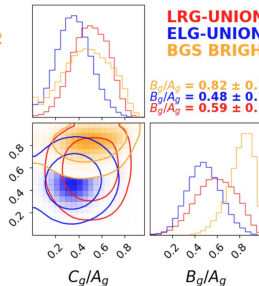


Galaxies

$$\begin{aligned} C_g/A_g &= 0.48 \pm 0.22 \\ C_g/A_g &= 0.36 \pm 0.15 \\ C_g/A_g &= 0.50 \pm 0.16 \end{aligned}$$

LRG-UNIONS $0.8 < z < 1.1$
ELG-UNIONS $0.8 < z < 1.6$
BGS BRIGHT-UNIONS $0.1 < z < 0.3$

$$\begin{aligned} B_g/A_g &= 0.82 \pm 0.11 \\ B_g/A_g &= 0.48 \pm 0.15 \\ B_g/A_g &= 0.59 \pm 0.19 \end{aligned}$$



Corinaldi et al. 2026, in prep

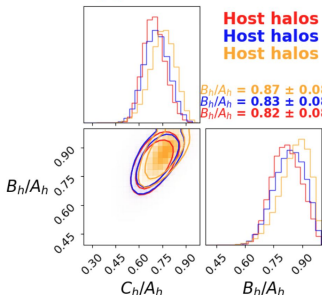
(2) ELG-LRG-BGS: constraints on 3D shapes

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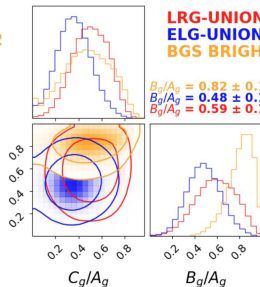


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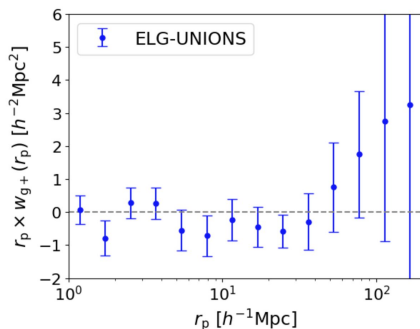
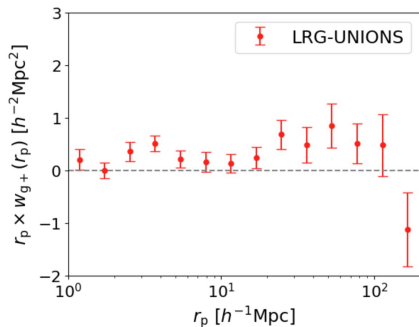
Corinaldi et al. 2026, in prep

ELG are oblate objects, LRG are prolate and BGS are a merge between both

Towards IA prediction with galaxy 3D morphology model

$$\xi_{g+}(r_p, \Pi) = \frac{S_+ D(r_p, \Pi) - S_+ R_D(r_p, \Pi)}{R_S R_D(r_p, \Pi)}$$

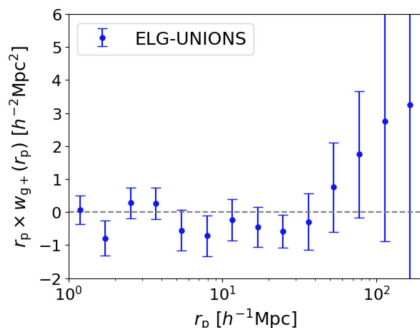
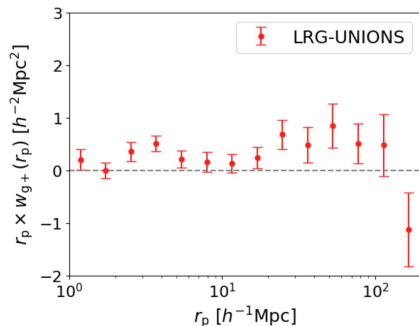
$$w_{g+}(r_p) = \int_{-\Pi_{\max}}^{\Pi_{\max}} d\Pi \xi_{g+}(r_p, \Pi)$$



Towards IA prediction with galaxy 3D morphology model

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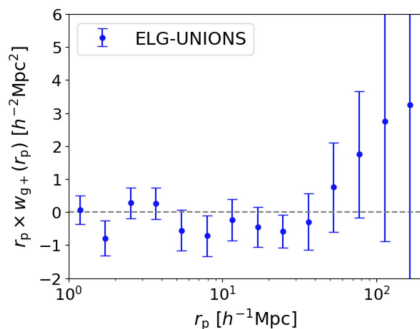
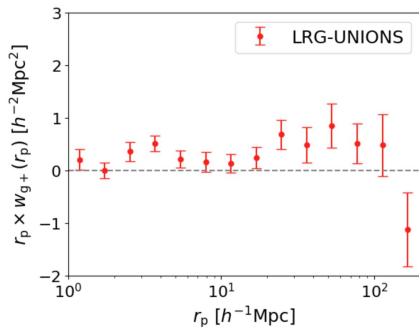


- **Future aim:** make a prediction of intrinsic alignment of galaxy using our model of 3D morphology

Towards IA prediction with galaxy 3D morphology model

$$\xi_{g+}(r_p, \Pi) = \frac{S_+ D(r_p, \Pi) - S_+ R_D(r_p, \Pi)}{R_S R_D(r_p, \Pi)}$$

$$w_{g+}(r_p) = \int_{-\Pi_{\max}}^{\Pi_{\max}} d\Pi \xi_{g+}(r_p, \Pi)$$



- **Future aim:** make a prediction of intrinsic alignment of galaxy using our model of 3D morphology
=> **motivation:** to reveal extra informations to intrinsic alignments about the 3D morphology of galaxies

Conclusion and next steps

- Development of a model of **3D morphology** for central galaxies inside dark matter halos using AbacusSummit N-body simulations

Conclusion and next steps

- Development of a model of **3D morphology** for central galaxies inside dark matter halos using AbacusSummit N-body simulations
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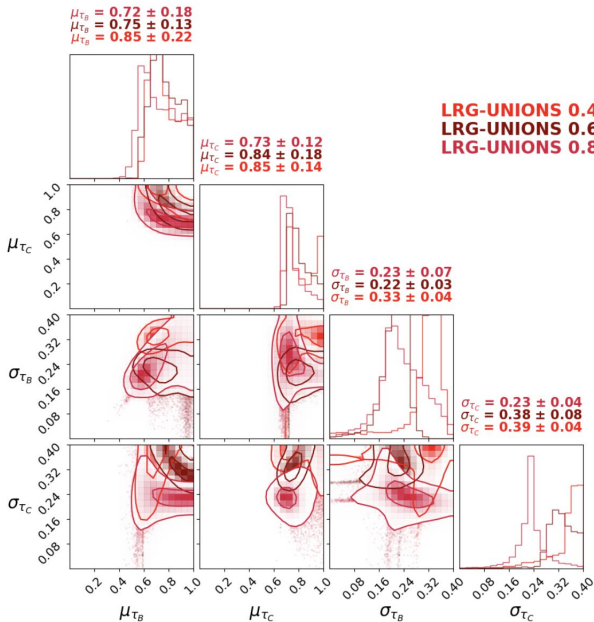
Conclusion and next steps

- Development of a model of **3D morphology** for central galaxies inside dark matter halos using AbacusSummit N-body simulations
- Development of a **simulation-based inference** method to measure the **distribution** of the 3D shapes of galaxies from the distribution of their **projected images**
- Ongoing works / next steps:
 - adding a **misalignment** parameter for the central galaxies
 - develop a model of 3D morphology for **satellite** galaxies
 - work with **hydrodynamic simulations**

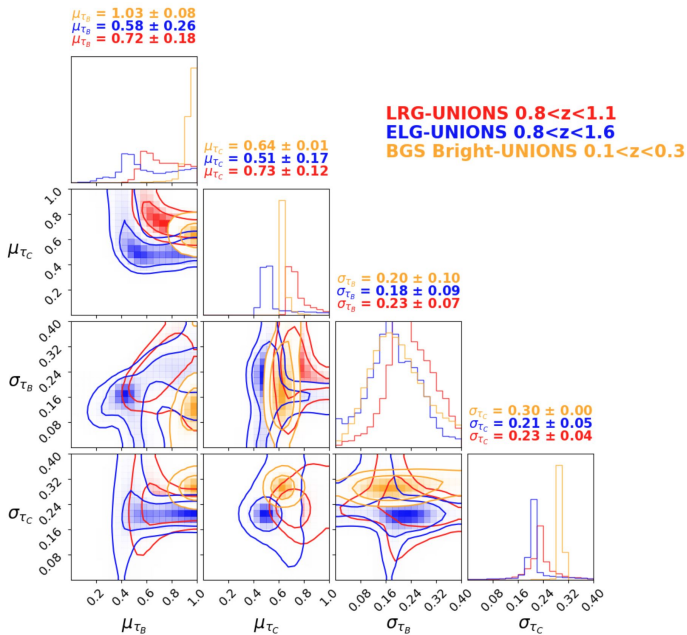
Thank you for your attention !

Back up slides

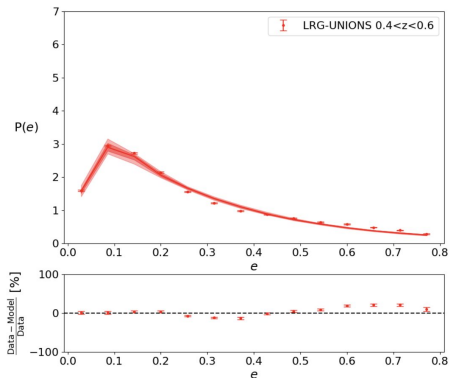
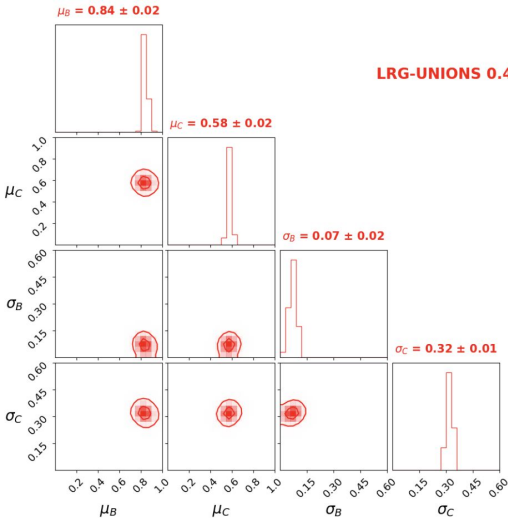
Redshift evolution: other parameter constraints



ELG-LRG-BGS: other parameter constraints



Toy-model for the LRG



Validation: coverage test for the LRG

LRG $0.4 < z < 0.6$

